Buffers and Regression Discontinuity

ECON 895 Spatial Techniques in Empirical Economics

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Plan for tonight

A boundary drawn in 1573

Constructive geometry

Building the running variable

The design

What it cost, and what it did not

We break around 8:45.

Two weeks of habits before we add more

- Count rows on both sides of every join, and then look at what you kept.
- `st_set_crs()` relabels. `st_transform()` moves. Know which you meant.
- Equal-area before `st_area()`. Units are a free check that catches only some of the errors.
- Plot it. Your eye caught the Indian Ocean faster than any summary would have.

Reminder: The Week 6 lab exam is closed book and closed model.

A boundary drawn in 1573

The mita

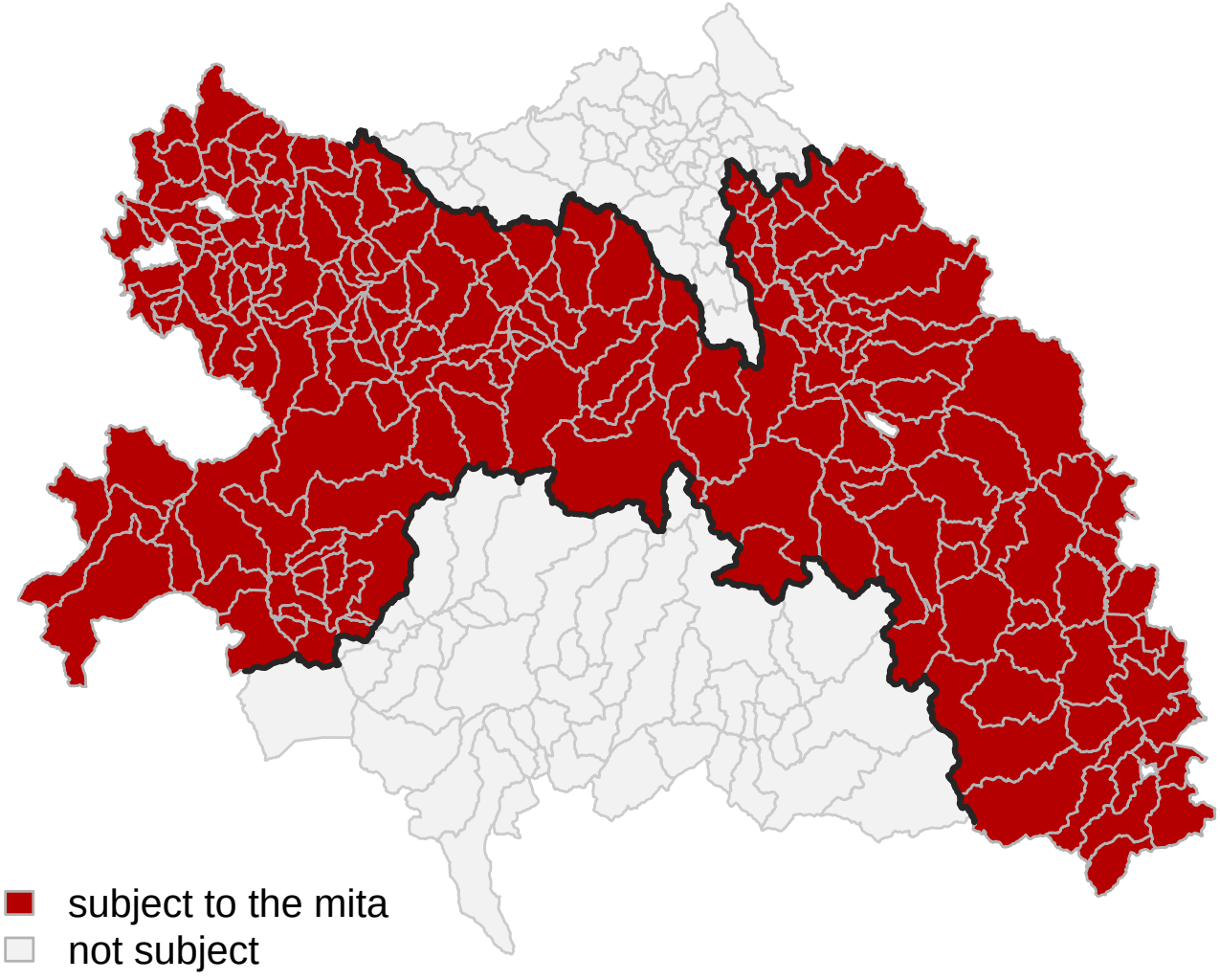
- From 1573 to 1812 the Spanish colonial state required subject communities in a defined region of Peru and Bolivia to send **one seventh of their adult men** to the silver mine at Potosí and the mercury mine at Huancavelica.

- The region was defined by a line. Inside it, the obligation. Outside it, nothing.

- The mita was abolished in 1812. The line has had no legal force for over two hundred years.

Dell (2010) asks whether you can still see the impact of this extractive institution in economic outcomes today.

The line, and who was inside it



205 of 299 districts in the study region were subject. The dark line is the boundary.

run it

What she finds

The mita is still visible in 2005

Children in districts that were subject to it are more likely to be stunted. Household consumption is lower. Two centuries after the institution ended.

- The obvious comparison is useless. Mita districts sit at different altitudes, on different soils, farther from different markets, with different colonial histories.

- Every one of those differences is a reason for a gap that has nothing to do with forced labor.

The idea that rescues it

Compare districts **just inside** the line with districts **just outside** it.

Regression discontinuity

A district five kilometers inside the boundary and another five kilometers outside sit on the same mountains, in the same climate, with the same distance to the same markets. What differs is that a colonial administrator put one on a list in 1573.

Hold that loosely. Dell (2010) names two criteria behind the line, distance to the mines and **elevation**, and her footnote 8 says the elevation constraint bound along the eastern and western boundaries, which follow the steep Andean precipice. She studies the segment where observables balance.

- The running variable is **distance to the boundary**. Positive inside, negative outside.

- All you have is a shapefile with a line in it. Getting from one to the other is what we're going to work on.

Constructive geometry

Buffers

Weeks 1 and 2 buffered points. A buffer on a polygon grows it and a **negative** distance shrinks it.

```
one <- d18[1, ]
c(original = as.numeric(st_area(one)),
  grown    = as.numeric(st_area(st_buffer(one,  5000))),
  shrunk   = as.numeric(st_area(st_buffer(one, -5000)))) / 1e6
```

```
##   original      grown     shrunk
## 288.129114 780.913970   2.763508
```

The inward buffer is how a boundary band gets drawn, and it is how a sample near a border gets defined. Note what a 5 km inward buffer does to a district that is only 288 square kilometers to begin with.

A centroid is not always in its own polygon

Of 299 Peruvian districts, how many have a centroid that falls outside the district itself?

```
cen <- st_centroid(st_geometry(d18))
pos <- st_point_on_surface(st_geometry(d18))

inside <- sapply(seq_len(nrow(d18)), function(i)
  st_within(cen[i], st_geometry(d18)[i], sparse = FALSE)[1, 1])
sum(!inside)                          # centroids outside their own district
```

```
## [1] 3
```

Three of 299, and Dell sidesteps it

A district shaped like a crescent or a horseshoe puts its center of mass in the gap. `st_point_on_surface()` is guaranteed to land inside a valid polygon. If a stray point crosses the boundary, that district switches treatment.

Name the predicate

A spatial relation is a question with a name. Say which one you asked.

```
c(cap_within  = sum(lengths(st_within(p18, mita_region)) > 0),
  cap_touch   = sum(lengths(st_intersects(p18, mita_region)) > 0),
  dist_within = sum(lengths(st_within(d18, mita_region)) > 0),
  dist_touch  = sum(lengths(st_intersects(d18, mita_region)) > 0))
```

```
##  cap_within   cap_touch dist_within  dist_touch
##         205         205         195         236
```

- On the capitals the two agree, because an interior point never lies on the line. On the districts they part by 41, and the answer is **three different things**.

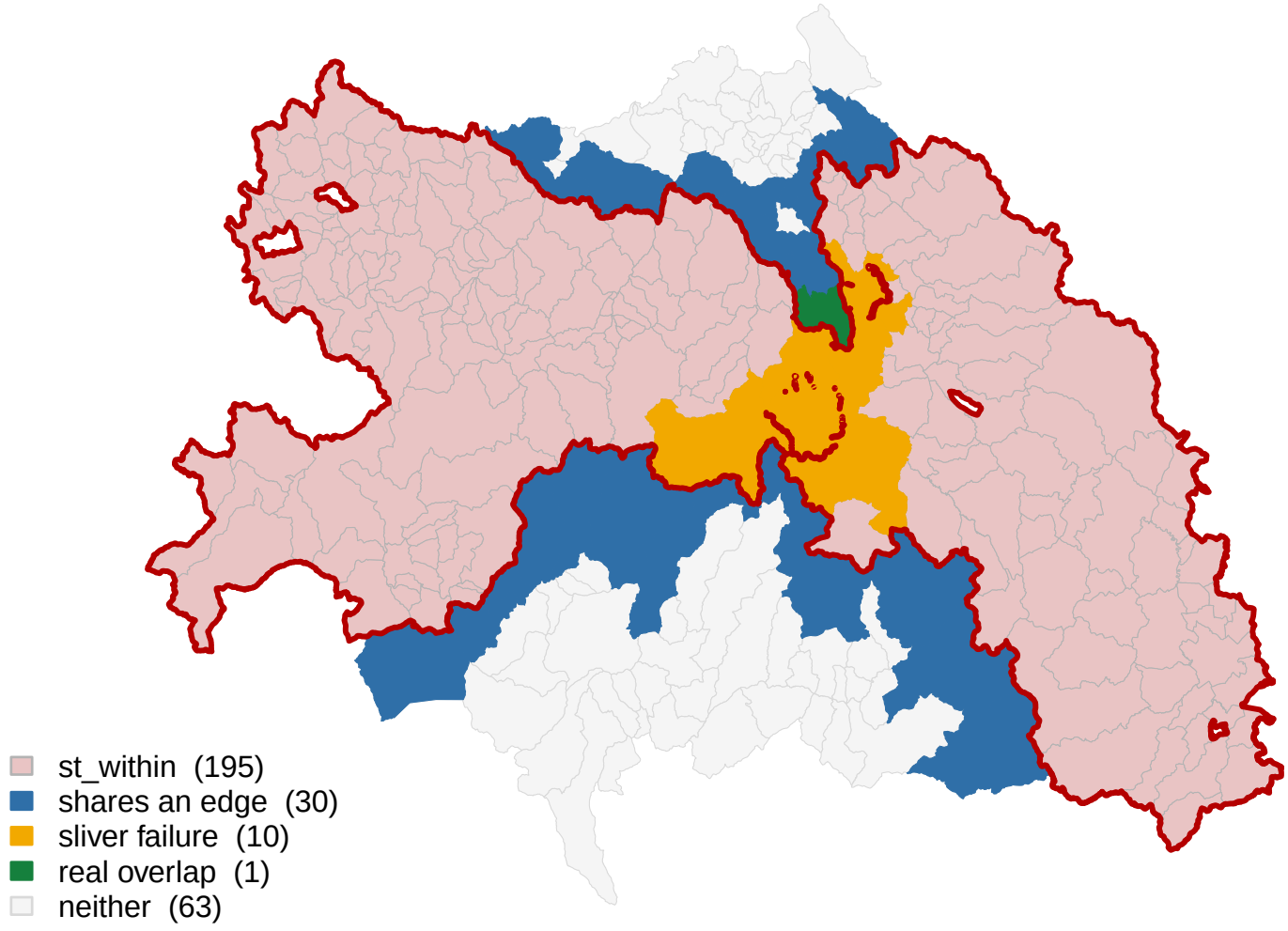
- 30 controls share an edge and overlap by exactly zero. `st_intersects` is true of a shared line and `st_within` is not. The predicates were asked different questions.

- 10 mita districts fail `st_within` against a dissolve of **themselves**, poking outside by around 10^{-8} square meters. Floating point.

- And one control overlaps the region by **2.1 percent of its own area**. Districts are supposed to partition, so that is an error in the shapefile, and only the count told you it was there.

run it

The gap is three different things



Blue is the honest difference between the predicates. Orange sits inside the red line and fails `st_within` anyway. Green is the one that should not exist. A join reported as "spatial" would have picked an answer for you and told you none of this.

run it

Clipping, and what it does quietly

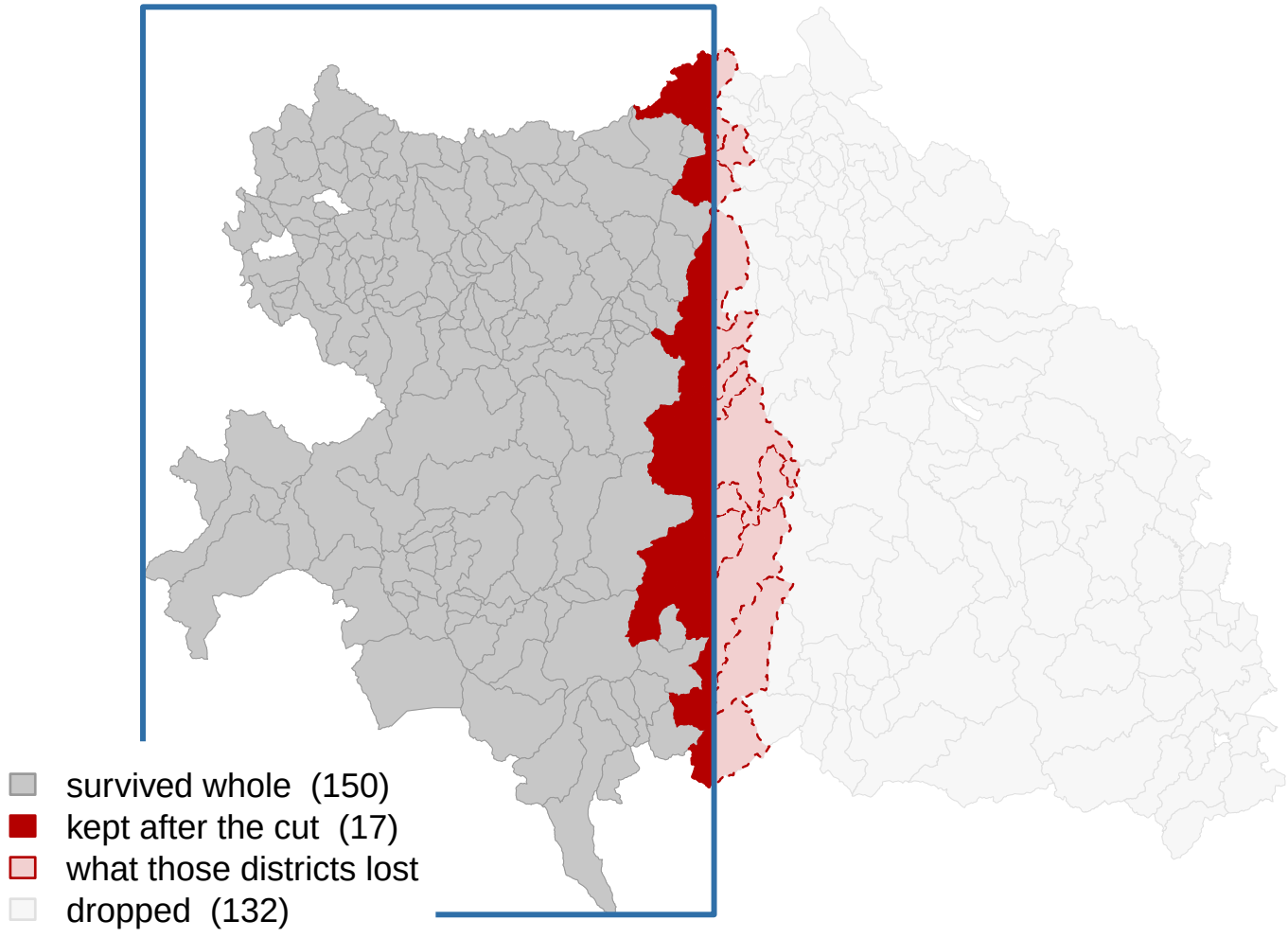
We keep only the western half of the study region.

```
bb  <- st_bbox(d18)
win <- st_as_sfc(st_bbox(c(xmin = unname(bb["xmin"]), ymin = unname(bb["
    ymin"]),
                           xmax = unname(mean(bb[c("xmin","xmax")])),
                           ymax = unname(bb["ymax"])), crs = st_crs(d18))
                          )

clipped <- st_intersection(d18, win)
nrow(d18); nrow(clipped)
```

299 districts become 167. That number is the easy part.

What the clip actually did



The dashed outline is what those districts looked like before. The straight edge is the window.

They kept their row, their code and their name, and lost the pale part.

run it

The district that kept its name and lost its land

17 districts survived the clip and came out smaller

The worst of them, `ubigeo` 40406, kept **1.3 percent** of its area and all of its identity. It still has its code, its name, and its row. Anything you compute per square kilometer after this is wrong for that district, and nothing warned you.

The habit, upgraded. Around a clip, count rows and compare areas. Week 1 said count rows. Polygons need the second half.

Simplifying breaks the seams

`st_simplify()` thins each polygon on its own, with no idea that its neighbor shares that edge.

```
s <- st_simplify(d18, dTolerance = 2000)
sum(st_is_empty(s))                                    # districts that
    vanished
```

```
## [1] 3
```

```
overlaps <- function(x) {                  # real overlap, not shared edges
  p <- suppressWarnings(st_intersection(x, x))
  p <- p[p$ubigeo != p$ubigeo.1, ]
  sum(as.numeric(st_area(p)) > 0) / 2
}
c(before = overlaps(d18), after = overlaps(s))
```

```
## before  after
##     11    565
```

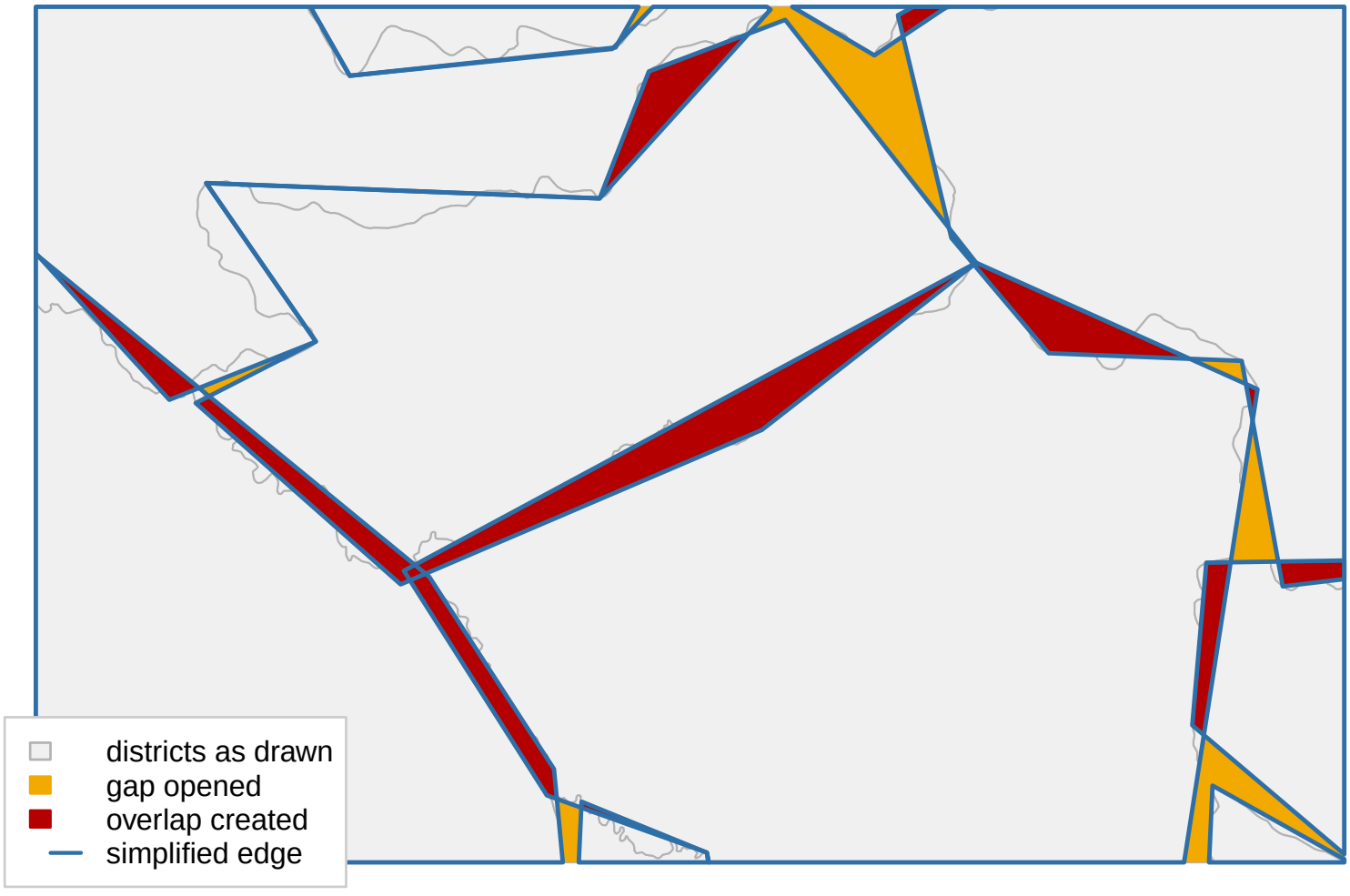
```
(sum(as.numeric(st_area(d18))) - sum(as.numeric(st_area(s)))) / 1e6
```

```
## [1] 1315.058
```

A district layer is a partition, so the before number ought to be zero. 3 districts vanish, the survivors overlap in 565 places, and 1,315 square kilometers of Peru go missing. Every survivor still passes `st_is_valid()`, because validity is a property of one polygon at a time and says nothing about the edges neighbors share.

run it

What simplification did to the seams



One seam, two neighbors, thinned separately. Where their new edges cross you get red, where they pull apart you get orange, and the alternating wedges run the length of every border in the layer. Each polygon is still valid on its own.

run it

The toolkit

- `st_buffer` draws bands around a boundary, and inward when the distance is negative.
- `st_centroid` and `st_point_on_surface` put a polygon on a map as a point. Dell needs one point per district.
- `st_within` and friends answer which side of the line something is on. That is the treatment indicator.
- `st_intersection` cuts geometries to a window, and quietly changes areas while keeping identities. Count rows and compare areas around it.
- `st_union` dissolves the 205 subject districts into one region, which is what `mita_region` is.
- `st_boundary` is the one you have not met, and it is the one the whole design turns on.

Break

Ten minutes.

Building the running variable

Before anything else, look at the coordinates

```
range(mita$lat)
```

```
## [1] 13.20417 15.93194
```

```
range(mita$lon)
```

```
## [1] 70.01667 74.23333
```

Peru is south of the equator and west of Greenwich

Both columns arrive as magnitudes with the sign stripped. Build points from them as they stand and your districts land in the **Arabian Sea**, off the west coast of India. This is Dell's own file, exactly as she distributes it, and it is the ordinary condition of archival data.

```
pts <- mita %>% mutate(lat = -abs(lat), lon = -abs(lon)) %>%
  st_as_sf(coords = c("lon", "lat"), crs = 4326, remove = FALSE)
```

How far is this district from the mita region?

```
inside_one <- p18 %>% filter(mita == 1) %>% slice(1)

st_distance(inside_one, mita_region)
```

```
## Units: [m]
##      [,1]
## [1,]    0
```

Zero

A polygon contains its interior. The distance from a point to a region it is inside is zero, which is the correct answer to the question you asked and useless as a running variable. The object you wanted is the edge.

st_boundary() is the whole trick

```
edge <- st_boundary(mita_region)

as.numeric(st_distance(inside_one, edge)) / 1000      # kilometers
```

```
## [1] 29.73395
```

st_boundary() casts a polygon down to the line around it. The same point, the same region, one function, and now a number you can regress on.

In practice you use the boundary file Dell (2010) ships, because the region's outer edge includes coastline and the far frontier, which are not the discontinuity.

Distance and a sign

Two questions. The first is how far. The second is which side.

```
pts$km     <- as.numeric(st_distance(p18, st_union(b18))) / 1000
pts$inside <- lengths(st_within(p18, mita_region)) > 0
pts$run    <- ifelse(pts$inside, pts$km, -pts$km)
```

```
sum(pts$inside == (mita$mita == 1))      # does st_within recover her
    dummy?
```

```
## [1] 299
```

299 of 299. We just rebuilt her treatment variable out of a shapefile and a predicate.

Now check it against hers

```
cor(pts$km, mita$dell_d_bnd)
```

```
## [1] 1
```

```
max(abs(pts$km - mita$dell_d_bnd)) * 1000            # meters
```

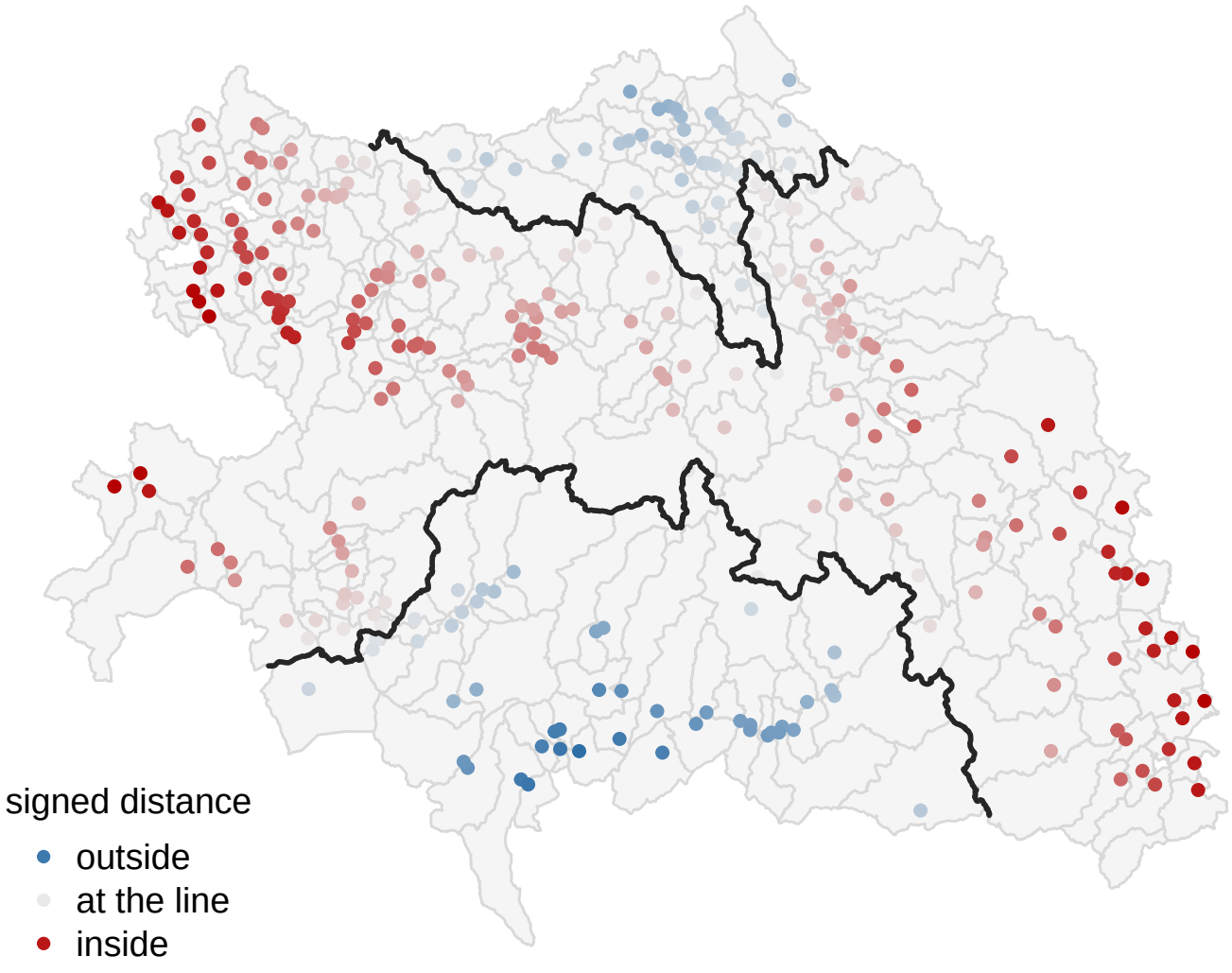
```
## [1] 0.400784
```

Correlation 1.00000, worst disagreement 40 centimeters

Her file ships `d_bnd` and we never looked at it until now. Rebuilding it from the shapefile and comparing is the whole method of this course in one slide.

Run the same thing in UTM **19**S rather than 18S and the worst disagreement becomes 326 meters. That is how you find out which zone she used. It does not make 18S correct. Zone 18S runs 78 to 72 degrees west and 119 of these 299 capitals sit east of that, where the scale factor runs a few tenths of a percent high. Reproduction and accuracy are different goals, and tonight we wanted reproduction.

Drawing the running variable



Red inside, blue outside, pale at the line. This is the variable the rest of the evening regresses on.

run it

The design

Writing it down

For district d, let Y_d be the share of children stunted, M_d the mita indicator, and x_d the signed distance we just built.

$$Y_d = \alpha + \underbrace{\gamma M_d}_{\text{the jump}} + \underbrace{f(|x_d|)}_{\text{the trend away from the line}} + X_d'\beta + \phi_b + \varepsilon_d$$

- $f(\cdot)$ is the RD polynomial. Dell runs it on $|x_d|$, the distance, so the trend is symmetric on the two sides and γ carries the whole level shift at the line.

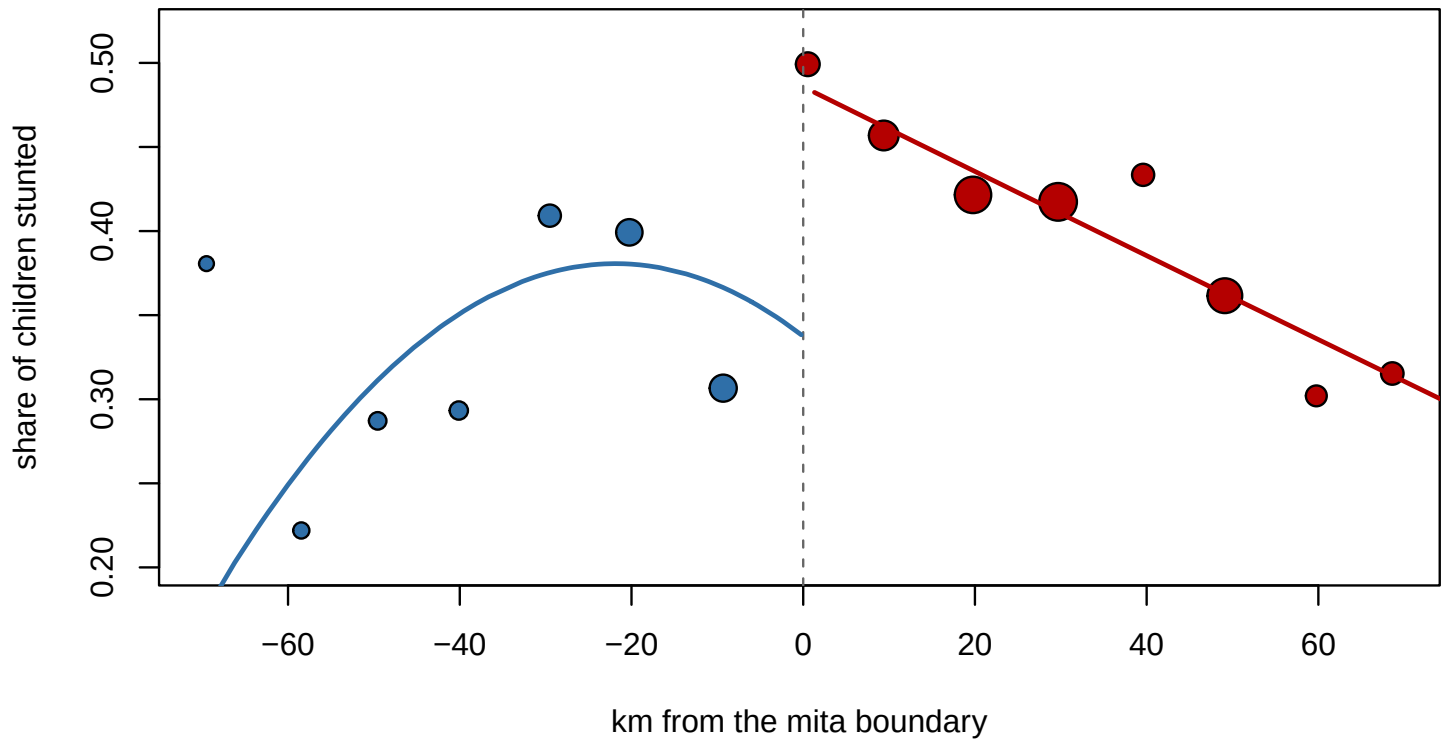
- X_d holds elevation and slope. ϕ_b is the fixed effect for boundary segment b, so districts are compared to others facing the same stretch of line. In the code they are `bfe4_1` to `bfe4_3`, three dummies for four segments.

- The identifying assumption is that everything except M_d varies **continuously** across the boundary. The assumption itself is not testable. Its implications are.

Look before you estimate

```
dd <- pts %>% st_drop_geometry() %>% filter(cusco != 1, !is.na(stunted))
c(before = nrow(pts), after = nrow(dd))                    # count both sides
```

```
## before  after
##    299    289
```



Binned means, ten kilometers per bin, dot size proportional to children. Zero is the boundary, positive is inside.

run it

The estimate

```
est <- dd %>% mutate(s = km / 100, s2 = s^2, s3 = s^3) %>%
  filter(km < 100)

m <- lm(stunted ~ mita + s + s2 + s3 + elevation + slope +
          bfe4_1 + bfe4_2 + bfe4_3, data = est, weights = n_children)

round(coeftest(m, vcov. = vcovHC(m, type = "HC1"))["mita", ], 3)
```

```
##    Estimate Std. Error    t value   Pr(>|t|)
##       0.073      0.023      3.171      0.002
```

Districts **just inside** the mita boundary have a stunting rate **7.3 percentage points** higher than districts just outside, on 158,848 children in 289 districts. Against a regional prevalence near 40 percent that is a relative increase of about a fifth, two centuries on, and it is a statement about the line rather than about a district deep inside the catchment. Dell (2010) Table II, Panel C, column 4 reports **0.073 (0.023)** on exactly that sample. All four numbers, from a shapefile and nine lines of R.

The polynomial runs on `km`, the distance. That is Dell's specification, and it constrains the trend to be symmetric on the two sides while the dummy carries the level shift. The signed variable gives the textbook form instead, which imposes a common slope across the line.

run it

Weighting: Something people rarely mention

```
mu <- lm(stunted ~ mita + s + s2 + s3 + elevation + slope +
           bfe4_1 + bfe4_2 + bfe4_3, data = est)      # no weights

c(weighted = unname(coef(m)["mita"]), unweighted = unname(coef(mu)["mita"
    ]))
```

```
##   weighted unweighted
## 0.07333885 0.04704053
```

0.073 weighted, 0.047 unweighted

A factor of 1.6, from a decision that appears nowhere in the equation. In the second, a district of two hundred children counts as much as one of nine thousand. That is a claim about what you are estimating.

Dell runs it on the children. Weighting each district average by its number of children reproduces that.

How wide is the window?

```
##       km estimate    se districts
## [1,] 100    0.073 0.023       289
## [2,]  75    0.061 0.022       239
## [3,]  50    0.064 0.024       185
```

All three rows are Dell (2010) Table II, Panel C, columns 4 to 6, and all three match. The estimate is stable, but each row is a different sample, so the three rows describe three populations rather than nested cuts of one. The bandwidth is an **assumption** about where the comparison is credible.

Does geography jump too?

```
##             estimate    se    p
## elevation      0.102 0.056 0.07
## slope         -2.199 0.593 0.00
```

Slope does not pass

Elevation is in thousands of meters and slope in degrees, so these are 102 meters and 2.2 degrees against a mean of 8.0. Terrain is measurably steeper on one side of a line drawn by a colonial administrator.

These are HC1. Dell (2010) reports Conley spatial standard errors for exactly these variables, roughly twice as wide, and on that basis calls elevation statistically identical across the boundary. Whether that is a repair or a concession is Week 9.

A predetermined covariate that jumps is evidence against continuity. Elevation and slope on the right-hand side assume a functional form for that imbalance rather than restoring the design.

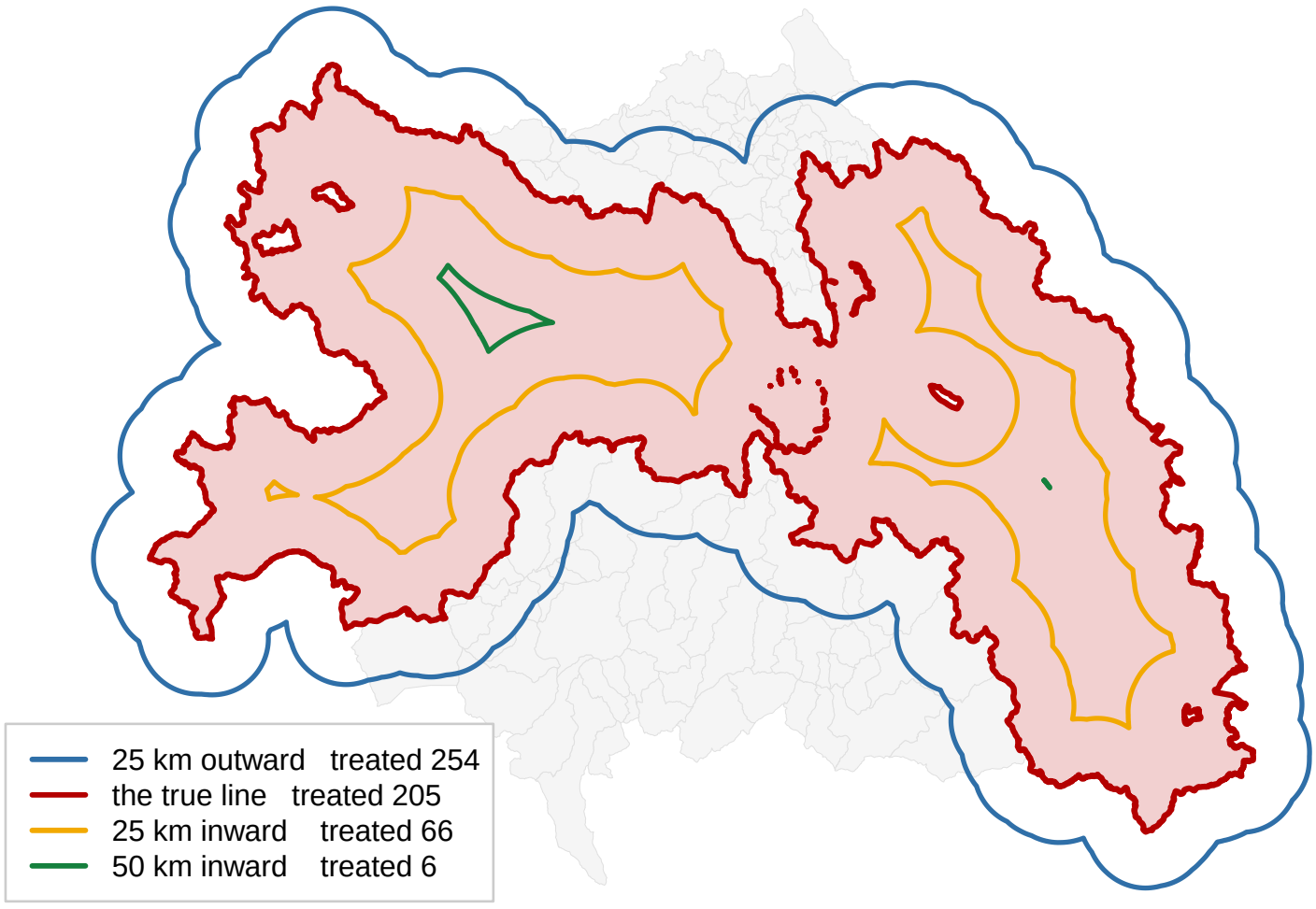
Move the line, and the effect should die

The negative buffer from the start of the evening builds the fake boundaries.

```
rd_at <- function(region) {
  km_new <- as.numeric(st_distance(p18, st_boundary(region))) / 1000
  t_new  <- as.integer(lengths(st_within(p18, region)) > 0)
  # New names on purpose. Write km = km[...] inside mutate() and the km
      on the
  # right is dd's OLD column, not this function's vector. Data masking.
  d <- dd %>% mutate(treat = t_new[match(ubigeo, pts$ubigeo)],
                     km    = km_new[match(ubigeo, pts$ubigeo)],
                     s = km/100, s2 = s^2, s3 = s^3) %>% filter(km < 100)
  mm <- lm(stunted ~ treat + s + s2 + s3 + elevation + slope +
             bfe4_1 + bfe4_2 + bfe4_3, data = d, weights = n_children)
  coeftest(mm, vcov. = vcovHC(mm, type = "HC1"))["treat", 1:2]
}
```

One function. Feed it the true region, then regions whose edge is a lie.

Where the fake lines actually go



Each fake line is a buffer of the real one, so every treated set is **nested** in the one outside it. Watch the count collapse. By 50 km inward there are 6 treated districts left, which is why the last rung of the ladder has almost no standard error to speak of.

run it

The placebo ladder

```
round(rbind(`true boundary`   = rd_at(mita_region),
            `25 km inward`    = rd_at(st_buffer(mita_region, -25000)),
            `25 km outward`   = rd_at(st_buffer(mita_region,  25000)),
            `50 km inward`    = rd_at(st_buffer(mita_region, -50000))),
                3)
```

```
##                Estimate Std. Error
## true boundary     0.057      0.029
## 25 km inward      0.049      0.029
## 25 km outward     0.030      0.034
## 50 km inward     -0.017      0.052
```

A buffer inside the real region gives a treated set nested in the true one, so attenuation is what both the null and the alternative predict. This is a dose-response check rather than a falsification test. The falsification you actually want moves the line sideways, onto a stretch of the Andes the mita never touched.

The true-line row reads 0.057 rather than the 0.073 from a few slides ago because this ladder measures distance to the region's own outer edge, which includes the frontier, whereas the estimate a few slides back used Dell's boundary file. Which edge you mean is never a detail.

How much does $f(\cdot)$ matter?

```
##                          estimate    se
## cubic in distance           0.073 0.023
## linear in distance          0.072 0.024
## cubic in lat and lon        0.070 0.044
## linear own slopes           0.082 0.055
## quadratic own slopes        0.140 0.084
```

Raising the **order** of f does nothing. Letting the slope **differ across the line** doubles the standard error and takes the result out of significance, and that is the shape modern practice reaches for first.

Gelman and Imbens (2019) argue global high-order polynomials should not be used in RD at all, because the weight they put on the cutoff is driven by observations far from it. Dell (2010) predates that consensus and reports every row above in her own Table III. The grown-up tool is `rdrobust`, which we are not installing tonight. Report the one you preregistered and show the others.

So do you believe it?

What survived tonight. Our running variable matches her published `d_bnd` to 40 centimeters, the estimate reproduces her Table II across three bandwidths, and it sits between 0.070 and 0.073 whether f is linear, cubic, or a nine-term surface in longitude and latitude.

What did not. Slope jumps at the boundary. Elevation was one of her two stated assignment criteria, so the line is correlated with terrain by construction, and terrain has its own claim on child growth. That is compound treatment, and it is the standing objection to every boundary design.

The honest verdict

Credible as a statement about the line, fragile as a statement about anything else. The reason to believe the paper is the second half you have not read yet, where she traces haciendas, roads, and market access. A balance table never carries a design on its own.

And the thing that would change my mind. Move the line sideways, onto a stretch of the Andes the mita never touched, and run the same regression. We did not do it tonight, and it is four lines of the geometry you now own.

What it cost, and what it did not

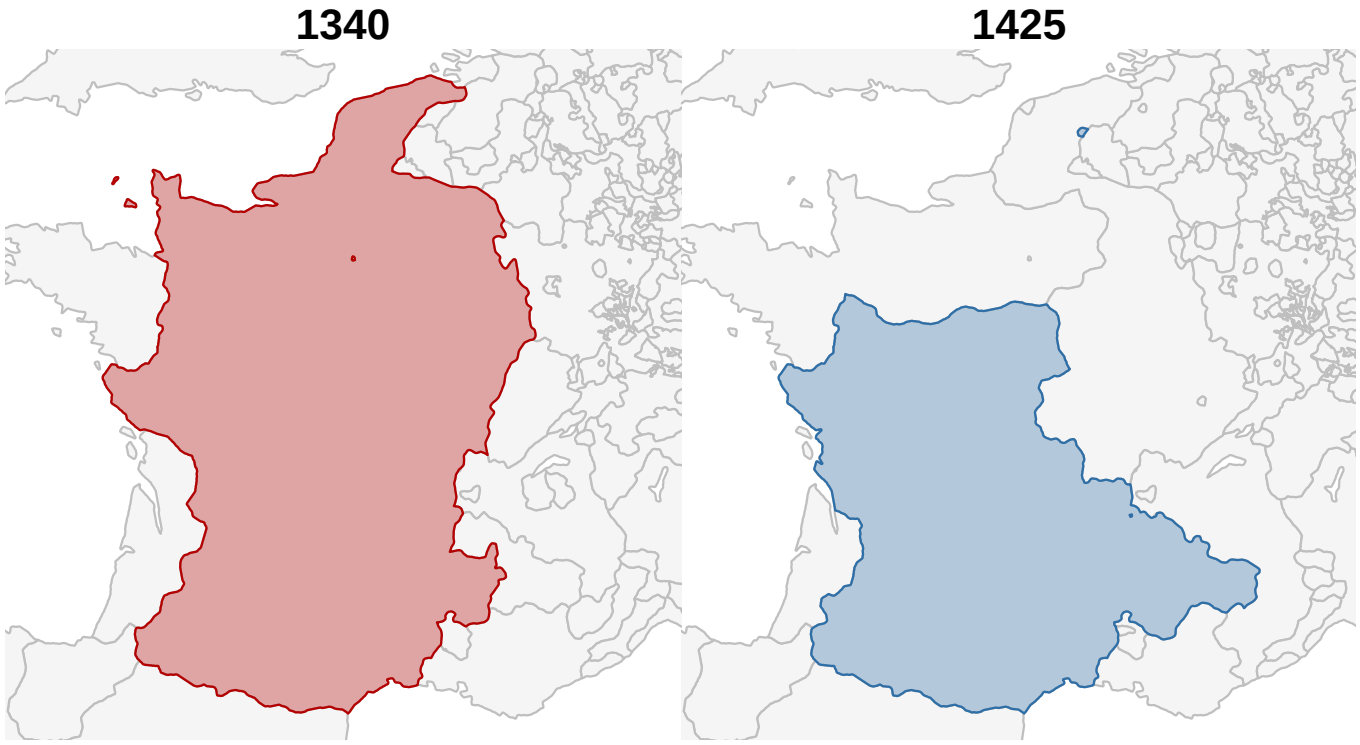
What it cost you to build

Three lines

`st_distance` to the boundary, `st_within` for the side, one `ifelse` for the sign. That is the running variable in a paper that has been cited thousands of times.

- A language model writes those three lines correctly, immediately, tonight.
- It cannot tell you whether a boundary drawn in 1573 assigned treatment as good as at random near the line, whether people sorted across it in the two centuries since, or whether the slope imbalance you just measured is fatal.
- That is the part you are being trained in, and it is the part that is scarce.

Boundaries that move



France in 1340 and again in 1425, across the first phase of the Hundred Years' War, drawn from Abramson's polity borders (Abramson, 2017). `st_sym_difference()` measures the change. **48.1 percent** of the larger footprint changed hands in 85 years. The mita line moved zero percent in 239 years.

When the boundary is the outcome

A project of mine asks whether the Black Death consolidated European states, again on Abramson (2017). There the boundary is the **dependent variable**.

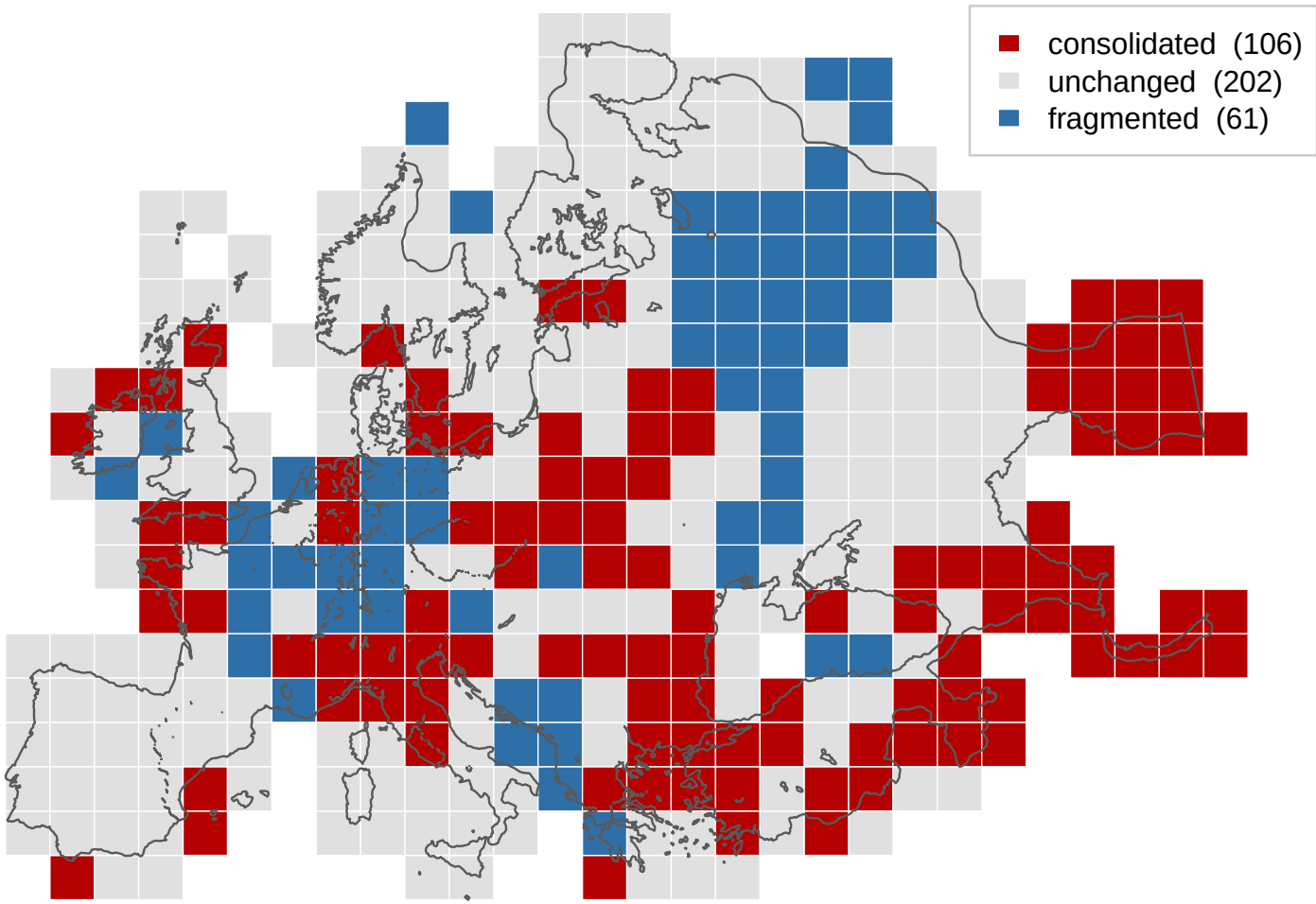
```
g <- st_make_grid(a1340, cellsize = 200000) %>% st_as_sf() %>%
  mutate(cell = row_number())
g$n1340 <- lengths(st_intersects(g, a1340))
g$n1425 <- lengths(st_intersects(g, a1425))

land <- g %>% filter(n1340 > 0 | n1425 > 0)
table(sign(land$n1425 - land$n1340))
```

```
##
##  -1   0   1
## 106 202  61
```

`st_make_grid()` lays a 200 km grid over Europe, a predicate counts polities per cell, and 369 cells carry at least one polity. 106 cells consolidated against 61 that fragmented. The project regresses that change on plague mortality, with pre-plague boundary change as the placebo, the same instinct as the ladder you just saw. Grids return next week as rasters, and interpolating the mortality surface is Week 8.

The outcome, drawn



The dependent variable, one cell at a time. Consolidation runs through the southeast and fragmentation through the north and the German lands, which is a pattern worth explaining rather than a number worth reporting. And 200 km is a choice. Halve the cell and the table changes, which is the modifiable areal unit problem wearing its research-design hat.

run it

Tonight in one slide

1. A polygon contains its interior, so distance to it is zero. `st_boundary()` is what you wanted.

2. Name your predicate. `st_within` recovered a published treatment variable exactly.

3. Around a clip, count rows and compare areas. Simplification breaks shared edges.

4. Weights and bandwidth are both assumptions. Weighting moved the answer by a factor of 1.6. The bandwidth happened not to, which is a result rather than a guarantee.

5. Rebuild the published variable and compare. Ours matched hers to 40 centimeters.

6. Move the line and the effect should attenuate. Ours did, and a nested fake line could not have done otherwise. Every standard error tonight is HC1, which assumes away spatial dependence in a design built entirely out of space. Week 9 is where we pay for that.

Before next week

For Week 4, Rasters and Difference-in-Differences

1. Read Nunn and Qian (2011) on the potato.
2. Work tonight's script by hand. Rebuild the running variable without looking at the slides.
3. Bring one boundary from your own field. We will ask whether it is a design.

Reminder. The Week 6 lab exam is closed book and closed model, and covers exactly this material.

References I

Abramson, S. F. (2017). The economic origins of the territorial state. *International Organization*, 71(1):97–130.

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